



© STEPHEN MATTHEW GRIFFIES ✧

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PRINCETON UNIVERSITY ATMOSPHERIC AND OCEANIC SCIENCES PROGRAM  
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RESEARCH AND TEACHING    ORCID    GOOGLE SCHOLAR    WEB OF SCIENCE

## RESEARCH STATEMENT

I pursue research questions related to geophysical fluid mechanics and the role of the physical ocean in the earth climate system. In pursuit of this research, I make use of theoretical concepts and mathematical methods, idealized process physics models, realistic numerical circulation models, and field measurements. Particular research topics include studies of Atlantic and Southern Ocean circulation dynamics; global and regional sea level; transport of matter and energy by mesoscale and submesoscale eddies; subgrid scale parameterizations of turbulent ocean stirring and mixing; analysis methods aimed at conceptually understanding ocean circulation and transport; physical and mathematical foundations of ocean circulation models; fundamentals of geophysical fluid mechanics.

## EDUCATION STATEMENT

As a teacher, mentor, author, and journal editor, I aim to foster an understanding of physical concepts and their creative and rigorous use in describing observed and simulated ocean and climate phenomena. I am particularly interested in revealing how concepts and methods from mathematical physics can be leveraged to deepen understanding of earth climate mechanics, and for fostering an appreciation of geophysical fluid mechanics within the broader context of theoretical mechanics and thermodynamics. Teaching and mentoring are central to all of my work, and I strive to generate curiosity and passion in colleagues and students while fostering an honest and non-judgmental scientific pursuit of how nature works.

## COLLABORATION STATEMENT

I nurture collaborations with scientists at all career stages who are passionate about furthering a rigorous understanding of ocean and climate physics, and with keen interests in mathematical physics.

## BROADER INTERESTS AND ACTIVITIES

meditation, yoga, walking, writing, sustainability, cultures, surfing, skiing

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## EDUCATION

|           |                                                                       |                                 |
|-----------|-----------------------------------------------------------------------|---------------------------------|
| 1995-1996 | Postdoctoral fellow in ocean and climate physics (mentor: Kirk Bryan) | Princeton University            |
| 1993-1995 | NOAA Climate and Global Change Fellow (mentor: Kirk Bryan)            | Princeton University            |
| 1988-1993 | PhD in theoretical high energy physics (advisor: Mirjam Cvetič)       | University of Pennsylvania      |
| 1987-1988 | pre-PhD studies in physics                                            | University of Washington        |
| 1986-1987 | Masters in engineering sciences & applied mathematics                 | Northwestern University         |
| 1981-1986 | Bachelor of science in chemical engineering                           | Louisiana State University      |
| 1978-1981 | High school (grades 10-12)                                            | Biloxi High School, Mississippi |

## SPECIAL TOPIC SCHOOLS

|          |                                                                          |                                  |
|----------|--------------------------------------------------------------------------|----------------------------------|
| Jan 1998 | NATO Advanced Study Institute: OCEAN MODELING AND PARAMETERIZATION       | Les Houches, France              |
| Feb 1995 | NATO Advanced Study Institute: CLIMATE VARIABILITY AND PREDICTABILITY    | Les Houches, France              |
| Jul 1994 | Meeting of UCAR Global and Climate Change Fellows                        | Steamboat Springs, Colorado, USA |
| Jul 1992 | Theoretical Advanced Study Institute: FROM STRING THEORY TO BLACK HOLES  | Boulder, Colorado, USA           |
| Jul 1991 | High Energy Physics and Cosmology School, Center for Theoretical Physics | Trieste, Italy                   |
| Jun 1991 | Theoretical Physics Summer School: PARTICLE PHYSICS IN THE 1990's        | Les Houches, France              |

## EMPLOYMENT AND APPOINTMENTS

|              |                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------|
| 2026–present | CNRS Research Director, IPSL/LOCEAN (Choose France for Science Laureate), Paris, France               |
| 2026–present | Research Scholar, Princeton University’s Atmospheric and Oceanic Sciences Program                     |
| 2025         | Lecturer with rank of professor, Princeton University’s Atmospheric and Oceanic Sciences Program      |
| 2024–2025    | Chair of the Visiting Scientist Committee for the Cooperative Institute for Modeling the Earth System |
| 2024–2025    | Director of Graduate Studies for Princeton University’s Atmospheric and Oceanic Sciences Program      |
| 2022–2024    | Graduate Work Committee for Princeton University’s Atmospheric and Oceanic Sciences Program           |
| 2020–2024    | Team lead for the GFDL Ocean/Cryosphere Division’s high resolution climate model project CM4X         |
| 2015–present | Lecturer in Princeton University’s Atmospheric and Oceanic Sciences Program                           |
| 2013–2017    | Member, GFDL Model Development Team Steering Committee                                                |
| Jun-Aug 2012 | Visiting Scientist, National Center for Atmospheric Research, Boulder, USA                            |
| Jan-Jun 2011 | Distinguished Visiting Scientist Fellow, CSIRO Marine and Atmospheric Research, Hobart, Australia     |
| Mar 2009     | Visiting Professor, Universite catholique de Louvain, Belgium                                         |
| Jan-Nov 2005 | Visiting Scientist, CSIRO Marine and Atmospheric Research, Hobart, Australia                          |
| 2001–2005    | Leader of the GFDL Oceans and Climate Group                                                           |
| 2000–2011    | Co-lead of the GFDL Ocean Model and Climate Model Development Teams                                   |
| 1996–2025    | Staff Physical Scientist, NOAA/GFDL (senior scientist 2011–2025, retired 1 May 2025)                  |
| 1995–1996    | Visiting Scientist, GFDL and Princeton University                                                     |
| 1993–1995    | NOAA Climate & Global Change Postdoc Fellow at Princeton University                                   |
| 1988–1993    | Physics Graduate Research Fellow, University of Pennsylvania Physics Department                       |
| 1986–1987    | Engineering Sciences and Applied Mathematics Fellow, Northwestern University                          |
| 1984–1986    | Chemical Engineering Research Laboratory Technician, Louisiana State University                       |

## AWARDS AND HONORS

|           |                                                                                                                                                                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026–2029 | Choose France for Science Laureate awarded through CNRS at IPSL/LOCEAN, Paris                                                                                                                                                                                                 |
| 2022      | NOAA Administrator’s Award (with 26 others) “For advancing the understanding of the Earth System by developing and applying NOAA’s state-of-the-art Coupled Carbon-Chemistry-Climate model”                                                                                   |
| 2021      | <a href="#">Reuters Hot List of Climate Scientists (#585)</a>                                                                                                                                                                                                                 |
| 2019      | Department of Commerce Silver Medal Award (with Robert Hallberg and Matthew Harrison): “For developing the state-of-the-art Modular Ocean Model version 6 (MOM6) to strengthen the Nation’s longer-range environmental prediction capabilities.”                              |
| 2019      | Sigma Xi scientific honor society                                                                                                                                                                                                                                             |
| various   | Web of Sciences (Clarivate) Highly Cited Researcher (2018, 2020, 2021, 2022, 2023, 2024)                                                                                                                                                                                      |
| 2017      | <a href="#">Elected Fellow of the American Geophysical Union</a> “For exceptional and sustained contributions to the understanding of large-scale ocean circulation and physics and seminal advances in ocean modeling”                                                       |
| 2017      | NOAA Administrator’s Award (with Robert Hallberg) “For scientific leadership for the innovation of the versatile community-based Modular Ocean Model MOM6”                                                                                                                    |
| 2014      | <a href="#">European Geosciences Union Fridtjof Nansen Medal for Oceanographic Research</a> “For outstanding contribution and leadership in ocean general circulation model development and critical insights in the physical nature and parameterization of ocean processes” |
| 2013      | Department of Commerce Silver Medal Award (with nine other GFDL staff scientists): “For development and application of NOAA’s first comprehensive Earth System Model that couples the carbon cycle and climate for projection of changes”                                     |
| 2012      | NOAA Administrator’s Award “For scientific vision, leadership and development of the Modular Ocean Model (MOM4) for climate modeling, research and predictions”                                                                                                               |
| 2011      | CSIRO Distinguished Visiting Scientist Fellow, Australia                                                                                                                                                                                                                      |
| 2009      | Visiting Professor, Universite catholique de Louvain, Belgium                                                                                                                                                                                                                 |
| 2001      | NOAA/Oceanic and Atmospheric Research Outstanding Scientific Review Paper                                                                                                                                                                                                     |
| 1999      | NOAA/Oceanic and Atmospheric Research Outstanding Scientific Paper                                                                                                                                                                                                            |
| 1998      | NOAA/Oceanic and Atmospheric Research Employee of the Year                                                                                                                                                                                                                    |
| 1997      | NOAA/Environmental Research Laboratories Outstanding Scientific Paper                                                                                                                                                                                                         |
| 1993–1995 | NOAA Climate and Global Change Research Fellow                                                                                                                                                                                                                                |

## PROFESSIONAL SERVICES AND MEMBERSHIPS

|              |                                                                                              |
|--------------|----------------------------------------------------------------------------------------------|
| 2026–present | Co-lead of the NEMO Diagnostics Working Group                                                |
| 2024–present | Scientific advisory board for the C-Star program at [C]Worthy                                |
| 2021–2025    | NEMO Scientific Advisory Committee                                                           |
| 2021–2026    | Editor-in-Chief for AGU's Journal of Advances in Modeling Earth Systems (JAMES)              |
| 2018–2020    | Ocean/Cryosphere Editor for AGU's Journal of Advances in Modeling the Earth System (JAMES)   |
| 2019–2023    | Chair of the awards committee for the EGU Fridtjof Nansen Medal for Oceanographic Excellence |
| 2017–2021    | Advisory board for the TICTOC Project, United Kingdom                                        |
| 2016–2019    | Awards committee for the EGU Fridtjof Nansen Medal for Oceanographic Excellence              |
| 2014–2018    | WCRP/CLIVAR Scientific Steering Group                                                        |
| 2014–2016    | Co-lead of the NCEP Climate Model Development Task Force                                     |
| 2012–2014    | CLIVAR/CLiC/SCAR Southern Ocean Region Implementation Panel                                  |
| 2012–present | WCRP/CLIVAR Ocean Model Development Panel (ex officio)                                       |
| 2010–present | European Geosciences Union (EGU)                                                             |
| 2009–2015    | Scientific Advisory Board for the Catalan Climate Institute IC3, Barcelona, Spain            |
| 2007–2018    | Editor of the journal <b>Ocean Modelling</b>                                                 |
| 2006–2009    | WCRP/CLIVAR Scientific Steering Group (ex officio)                                           |
| 2004–2009    | WCRP/CLIVAR Working Group on Coupled Modelling (ex officio)                                  |
| 2004–2007    | Editorial Board of the journal <b>Ocean Science</b>                                          |
| 1999–2012    | WCRP/CLIVAR Working Group on Ocean Model Development (co-chair 2004–2009)                    |
| 1993–present | American Geophysical Union and American Meteorological Society                               |

## UNIVERSITY TEACHING

- Autumn semester 2025: Princeton University Atmospheric and Oceanic Sciences 585: Special topics: **Methods in the Analysis of Ocean Scalar Properties**. 24 lectures of 80 minutes duration covering the full course, presenting a suite of theoretical tools to support physical ocean analysis. Topics include material tracers, tracer advection and diffusion, turbulent tracer transport, subgrid scale parameterizations, Green's function methods, watermass transformation analysis. Lecture material based on [Griffies \(2026\): Geophysical Fluid Mechanics](#).
- Spring semester 2023, 2024, 2025: Princeton University Atmospheric and Oceanic Sciences 572: **Theory of Geophysical Fluid Waves and Instabilities**. 24 lectures of 80–90 minutes duration covering the full course, presenting theoretical foundations for ocean and atmosphere linear wave mechanics and instability theory. Topics include wave kinematics, acoustics, capillary waves, surface gravity waves, inertial waves, Rossby waves, shallow water waves, internal inertia-gravity waves, shear instability, inertial instability, symmetric instability, baroclinic instability. Lecture material based on [Griffies \(2026\): Geophysical Fluid Mechanics](#).
- Autumn semester 2014 (0.5), 2015 (0.5), 2016 (0.5), 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024 (0.5): Princeton University Atmospheric and Oceanic Sciences 571: **Foundations of Geophysical Fluid Mechanics**. 24 (12) lectures of 80–90 minutes duration covering the full (half) of course, focusing on the theoretical foundations for geophysical fluid mechanics. Topics include mechanics of motion on a rotating planet, Eulerian and Lagrangian fluid kinematics, frictional stresses, pressure and form stress, buoyancy, Ekman mechanics, shallow water mechanics, vorticity and potential vorticity, quasi-geostrophy. Lecture material based on [Griffies \(2026\): Geophysical Fluid Mechanics](#).
- Spring semester 2020: Princeton University Atmospheric and Oceanic Sciences 521: **Southern Ocean Seminar**. I provided five 90-minute lectures covering Southern Ocean dynamics, while other lecturers presented allied material to cover Southern Ocean science. Lecture material based on [Griffies \(2026\): Geophysical Fluid Mechanics](#).
- Spring semester 2017, 2018, 2019, 2024: Princeton University Atmospheric and Oceanic Sciences 580: **Special Topics on Great Papers in Atmospheric and Oceanic Sciences**. I led on 90-minute discussion session on a chosen classic paper in ocean fluid mechanics.
- Spring semester 2016, 2019: Princeton University Geosciences 503: **Responsible Conduct of Research in Geosciences**. I co-taught one three-hour discussion session on ethical behavior in research.
- Autumn semester 1993: Princeton University Atmospheric and Oceanic Sciences Special Topics 580: **Data Assimilation in Atmospheric and Oceanic Models**. I was the co-lecturer and coordinator of visiting lectures throughout the semester.
- 1990–1993: Instructor, Undergraduate Physics Laboratory, University of Pennsylvania
- 1990–1993: Teaching Assistant, General Relativity and Quantum Field Theory, University of Pennsylvania

## MENTORING, PHD COMMITTEES, AND SABBATICAL HOSTING

|              |                         |                                                                        |
|--------------|-------------------------|------------------------------------------------------------------------|
| 2027-2029    | Jamie Hilditch          | CNRS/LOCEAN postdoc                                                    |
| 2027-2029    | Oliver Street           | CNRS/LOCEAN postdoc                                                    |
| 2025-present | Hunter Camp             | Princeton University graduate student (AOS committee member)           |
| 2024         | Claire Yung             | Visiting graduate student (from ANU, Canberra, AUS)                    |
| 2024-present | Stefan Kildal-Brandt    | Princeton University graduate student (geophysics committee member)    |
| 2023-present | Kiera Lowman            | Princeton University graduate student (AOS committee member)           |
| 2023-present | Maxime Keutgen De Greef | Princeton University graduate student (AOS committee member)           |
| 2023-present | Kentaro Hanson          | Princeton University graduate student (applied maths committee member) |
| 2023         | Jan Zika                | Princeton University visiting scholar (from UNSW, Sydney, AUS)         |
| 2022-present | Matthew Lobo            | Princeton University graduate student (AOS, primary mentor)            |
| 2022-present | Winnie Chu              | Princeton University graduate student (AOS committee member)           |
| 2022-2025    | Wenda Zhang             | Princeton University postdoc fellow                                    |
| 2021-2022    | Rachel Pang             | Princeton University undergraduate (junior paper mentor)               |
| 2021         | Abigail Bodner          | Brown University graduate student (PhD thesis reader)                  |
| 2020-2025    | Jan-Erik Tesdal         | Princeton University postdoc fellow                                    |
| 2020         | Ruth Moorman            | Princeton University predoc intern                                     |
| 2019-2020    | Benjamin Taylor         | Princeton University predoc intern                                     |
| 2019-2021    | Hemant Khatri           | Princeton University postdoc fellow                                    |
| 2019-2020    | Elizabeth Yankovsky     | Princeton University graduate student (AOS committee member)           |
| 2019         | Hussein Aluie           | Princeton University visiting scholar (from University of Rochester)   |
| 2018-2022    | Graeme MacGilchrist     | Princeton University postdoc fellow                                    |
| 2017-2022    | Houssam Yassin          | Princeton University graduate student (AOS, primary mentor)            |
| 2017-2018    | Laure Zanna             | Princeton University visiting scholar (from Oxford University)         |
| 2017         | Jianjun Yin             | Princeton University visiting scholar (from University of Arizona)     |
| 2016-2019    | Brandon Reichl          | Princeton University postdoc fellow                                    |
| 2016-2018    | Nathaniel Tarshish      | Princeton University predoc intern                                     |
| 2015-2017    | Amanda O'Rourke         | University of Michigan postdoc fellow (with Brian Arbic)               |
| 2015-2016    | Henri Drake             | Princeton University predoc intern                                     |
| 2014-2017    | Anna FitzMaurice        | Princeton University PhD student (AOS committee member)                |
| 2014-2015    | Ivy Frenger             | Princeton University postdoc fellow (with Jorge Sarmiento)             |
| 2014         | Magnus Hieronymus       | Stockholm University graduate student (PhD opponent)                   |
| 2013-2017    | Robert Nazarian         | Princeton University PhD student (AOS committee member)                |
| 2013-2016    | Adele Morrison          | Princeton University postdoc fellow (with Jorge Sarmiento)             |
| 2013         | Terrence O'Kane         | GFDL visiting scholar from CSIRO Marine Research, Hobart, Australia    |
| 2012-2017    | Carolina Dufour         | Princeton University postdoc fellow (with Jorge Sarmiento)             |
| 2012-2013    | Yalin Fan               | Princeton University postdoc fellow                                    |
| 2011-2014    | Michael Bueti           | University of Rhode Island PhD student (committee member)              |
| 2008-2011    | Michael Bates           | University of New South Wales PhD student (committee member)           |
| 2005-2009    | Andreas Klocker         | University of Tasmania PhD student (committee member)                  |
| 2003-2004    | Rüdiger Gerdes          | GFDL visiting scholar (from AWI, Bremerhaven, Germany)                 |
| 2001-2002    | Harper Simmons          | GFDL postdoc fellow                                                    |
| 1999-2002    | Shafer Smith            | Princeton University postdoc fellow                                    |

## OCEANOGRAPHIC FIELD WORK

- Mar-May 2017: Eight week cruise on the *RRS JC Ross* to the Orkney Passage and Scotia Sea, as part of the Dynamics of the Orkney Passage Outflow (DynOPO) project. Principal Scientific Officer: Alberto Naveira Garabato.
- Jul 1993: Two week cruise on the *CCGS Hudson* to the Labrador Sea in support of the WOCE Line AR7W Atlantic Circulation Experiment. Chief Scientist: John Lazier.

## PARTICIPANT/COLLABORATOR ON RESEARCH GRANTS AND PROJECTS

- Partner Investigator (2026-2032) for the Australian Research Council (ARC) Centre of Excellence in Our Future Oceans, AU\$44M.
- Partner Investigator for the Australian Research Council (ARC) Centre of Excellence in Antarctic Science (ACEAS) (2021-2025), AU\$25M.

- PI for an NSF subcontract with New York University for NOAA's Climate Variability and Predictability Program Climate Process Team: *Ocean Transport and Eddy Energy* (2024-2025), \$150K.  
*Diagnostics and Performance Metrics for Evaluating Ventilation Pathways and Interior Water Mass Properties in Ocean Models* (2020-2022), \$180K.
- Co-PI for NOAA's Climate Variability and Predictability Program Climate Process Team: *Ocean Transport and Eddy Energy* (2020-2024), \$770K.
- Co-PI for NOAA's Climate Variability and Predictability Program (CVP) *Decadal Climate Variability and Predictability* proposal *Drivers of coastal sea level change along the eastern US* (2020-2023), \$200K.
- PI for DOE subcontract with Princeton University for the *Diagnostics and Performance Metrics for Evaluating Ventilation Pathways and Interior Water Mass Properties in Ocean Models* (2020-2022), \$180K.
- Co-PI for Australian Research Council Discovery Project (2019-2022): *Risks of rapid ocean warming at the Antarctic continental margin*. AU\$582K.
- Co-PI for NOAA Modeling, Analysis, Predictions, and Projections Program (01Aug2018–31Jul2020): *Addressing Key Issues in CMIP6-era Earth System Models*. \$434K.
- Program advisory board for the UK NERC funded project: *Transient tracer-based Investigation of Circulation and Thermal Ocean Change (TICTOC)* (2017-2021).
- Partner Investigator for the Australian Research Council (2017-2023) Centre of Excellence for Climate Extremes, AU\$30M.
- Co-PI for the Ocean Model Intercomparison Project (OMIP), which is part of the Coupled Model Intercomparison Project (CMIP6) (2016-2022).
- Co-PI for the Flux Anomaly Forcing Model Intercomparison Project (FAFMIP), which is part of the Coupled Model Intercomparison Project (CMIP6) (2016-2022).
- Co-PI for NOAA Modeling, Analysis, Predictions, and Projections Program (01Jul2016–30Jun2018): *Development toward NCEP's fully-coupled global forecast and data assimilation system: A coupled wave-ocean system*. \$316K.
- Partner Investigator for the Australian Research Council (2016-2020) funded project: *An Australian Consortium for Eddy-Resolving Ocean-Sea Ice Modelling*, AU\$599K.
- US Department of Energy (15Aug2014–14Aug2017): *Three-dimensional structure of the Southern Ocean overturning circulation*, \$624K.
- US National Science Foundation (01Sep2014–31Aug2020): *Southern Ocean Carbon and Climate Observations and Modeling (SOCCOM)*, \$20.9M
- NASA (26Jun2014–25 Jun2017): *The role of mesoscale eddies in cross-frontal transport and subduction of nutrients and carbon in the Southern Ocean*, \$715K.
- NOAA (01Sept2013–31Aug2016): *Signature of the Atlantic meridional overturning circulation in the North Atlantic dynamic sea level*, \$393K.
- US Department of Energy (15Sep2011–14Sep2015): *Mode and intermediate waters in Earth System Models*, \$519K.
- Partner Investigator for the Australian Research Council (2011-2018) Centre of Excellence for Climate System Science, AU\$21.4M.
- NOAA Climate Program Office and US National Science Foundation (2010–2015): *Climate Processes Team on representing internal-wave driven mixing in global ocean models*.
- NOAA Climate Program Office and US National Science Foundation (2003–2008): *Climate Processes Team on ocean eddy mixed layer interactions*.
- NOAA Climate Program Office and US National Science Foundation (2003–2008): *Climate Processes Team on gravity current entrainment*.

## CONVENER/ORGANIZER OF WORKSHOPS & MEETINGS

- Oct 2021: Scientific advisory committee for the WCRP/CLIVAR workshop: Future Directions in Basin and Global High-resolution Ocean Modelling, GEOMAR, Kiel, Germany (virtual).
- Mar 2019: Scientific advisory committee for the WCRP/CLIVAR workshop: Sources and sinks of ocean mesoscale eddy energy, Florida State University, Tallahassee, Florida, USA.
- Feb 2018: Co-convenor for the Town Hall: Process understanding and standardized assessment towards the eddying realm. AMERICAN GEOPHYSICAL UNION OCEAN SCIENCES CONFERENCE, Portland, Oregon, USA.
- Feb 2018: Co-convenor for the session: Modeling the Climate System at High Resolution, AMERICAN GEOPHYSICAL UNION OCEAN SCIENCES CONFERENCE, Portland, Oregon, USA.
- Sep 2016: Science Organizing Committee and Executive Planning Team for CLIVAR OPEN SCIENCE CONFERENCE, Qingdao, China.
- Apr 2014: PHYSICAL AND BIOGEOCHEMICAL OCEAN MODELLING: DEVELOPMENT, ASSESSMENT, AND APPLICATIONS, Session at the European Geosciences Union General Assembly, Vienna, Austria.
- Feb 2014: PHYSICAL AND BIOGEOCHEMICAL OCEAN MODELING: DEVELOPMENT, ASSESSMENT AND APPLICATIONS, Session at the Ocean Sciences meeting, Honolulu, Hawaii.
- Apr 2013: PHYSICAL AND BIOGEOCHEMICAL OCEAN MODELLING: DEVELOPMENT, ASSESSMENT, AND APPLICATIONS, Session at the European Geosciences Union General Assembly, Vienna, Austria.
- Feb 2013: CLIVAR WGOMD/SOP WORKSHOP ON SEA-LEVEL RISE, OCEAN/ICE-SHELF INTERACTIONS, AND ICE SHEETS, Hobart, Australia.
- Apr 2012: PHYSICAL AND BIOGEOCHEMICAL OCEAN MODELLING: DEVELOPMENT, ASSESSMENT, AND APPLICATIONS, Session at the European Geosciences Union General Assembly, Vienna, Austria.
- Oct 2011: OCEAN CIRCULATION AND VENTILATION, Session at the WCRP Open Science Conference, Denver, USA.
- Apr 2011: PHYSICAL AND BIOGEOCHEMICAL OCEAN MODELLING: DEVELOPMENT, ASSESSMENT, AND APPLICATIONS, Session at the European Geosciences Union General Assembly, Vienna, Austria.
- Oct 2009: WORKSHOP ON OCEAN CLIMATE MODELING, GFDL/Princeton, USA.
- Apr 2009: CLIVAR WORKSHOP ON OCEAN MESOSCALE EDDIES: OBSERVATIONS, SIMULATIONS, AND PARAMETERIZATIONS, Exeter, UK.
- Aug 2007: CLIVAR WORKSHOP ON NUMERICAL METHODS IN OCEAN MODELLING, Bergen, Norway.
- Nov 2005: CLIVAR WORKSHOP ON MODELLING THE SOUTHERN OCEAN, Hobart, Australia.
- Jun 2004: CLIVAR WORKSHOP ON EVALUATING THE OCEAN COMPONENT OF IPCC MODELS, Princeton, USA.
- Aug 2002: WORKSHOP ON Z-COORDINATE OCEAN MODELING, Massachusetts Institute of Technology, USA.
- Nov 1999: MEETING OF Z-COORDINATE OCEAN MODELING AT GFDL, LANL, MIT, AND NCAR, Princeton, USA.
- Jul 1999: OCEAN/ATMOSPHERE VARIABILITY AND PREDICTABILITY, Session at the International Union of Geodesy and Geophysics, Birmingham, UK.

## RESEARCH PRESENTATIONS SINCE 2010

- Sept 2026: PROBLEMS AND PROSPECTS AT THE FRONTIER OF OCEAN CLIMATE MODEL THEORY AND PRACTICE, Virtual seminar to Imperial College, London.
- Jun 2026: CLIMATE MODEL THERMAL EQUILIBRATION IN AN OCEAN MESOSCALE-DOMINANT REGIME, seminar to LOCEAN, Paris.
- Mar 2025: CLIMATE MODEL THERMAL EQUILIBRATION IN AN OCEAN MESOSCALE DOMINANT REGIME, virtual talk to the Harvard University Earth and Planetary Sciences Department.
- Dec 2024: MOM6 AND THE CM4X CLIMATE MODEL HIERARCHY, talk given at the annual meeting of the American Geophysical Union, Washington, DC, USA.

- Oct 2024: MOM6 AND THE CM4X CLIMATE MODEL HIERARCHY, talk given at the Center for Ocean/Atmosphere Science, Courant Institute of Mathematical Sciences, New York University, USA.
- Sep 2024: MOM6 AND THE CM4X CLIMATE MODEL HIERARCHY, talk given at the COMMODORE workshop on numerical methods in ocean modeling, National Center of Atmospheric Research, Boulder, USA.
- Aug 2024: MOM6 AND THE CM4X CLIMATE MODEL HIERARCHY, talk given at the annual Eddy Energy and Transport Climate Process Team meeting, Brown University, USA.
- June 2024: MODELING THE OCEAN WITH MOM6: A PRIMER ON VERTICAL LAGRANGIAN REMAPPING AND THE ANALYSIS OF FINITE VOLUME OCEAN MODELS, virtual seminar given at Alfred Wegener Institute, Germany, in honor of the retirement of Rüdiger Gerdes.
- April 2024: MODELING THE OCEAN WITH MOM6: A PRIMER ON VERTICAL LAGRANGIAN REMAPPING AND THE ANALYSIS OF FINITE VOLUME OCEAN MODELS, virtual seminar given at University of Rhode Island Graduate School of Oceanography.
- Feb 2022: A MATHEMATICAL FORMALISM FOR CIRCULATION IN WATER MASS CONFIGURATION SPACE, virtual presentation at AGU Ocean Sciences Meeting.
- May 2020: THE IMPORTANCE OF REFINED MODEL RESOLUTION FOR UNDERSTANDING AND PROJECTING GLOBAL, REGIONAL, AND COASTAL SEA LEVEL, NASA GISS virtual Sea Level Seminar Series.
- Jan-April 2020: VERTICAL LAGRANGIAN-REMAPPING, GENERALIZED VERTICAL COORDINATES, AND SPURIOUS DIAPYCNAL MIXING IN OCEAN MODELS: COMMODORE meeting, Hamburg, Germany; DRAKKAR meeting, Grenoble, France; AGU Ocean Sciences, San Diego, USA; CESM Ocean Model Working Group.
- Oct 2019: WATER MASS TRANSFORMATION (WMT) ANALYSIS AND TRACER BUDGETS WITH GENERALIZED VERTICAL COORDINATES AND VERTICAL LAGRANGIAN-REMAPPING, annual meeting of the Transient tracer-based Investigation of Circulation and Thermal Ocean Change (TICTOC) Project, Exeter, UK.
- Feb 2019: WATER MASS TRANSFORMATION ANALYSIS IN OCEAN MODELS: SOME THOUGHTS AND QUESTIONS, workshop on Water mass transformation for ocean physics and biogeochemistry, University of New South Wales, Sydney, Australia.
- May 2018: UNDERSTANDING AND PROJECTING GLOBAL, REGIONAL, AND COASTAL SEA LEVEL: REASONS TO INCLUDE COASTAL OCEAN PROCESSES IN GLOBAL MODELS: Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA) Annual Meeting, Australian National University, Canberra, Australia & University of New South Wales, Sydney, Australia; ISSI workshop on understanding the relationship between coastal sea level and large-scale ocean circulation, Bern, Switzerland.
- Feb 2018: SUBSURFACE WARMING OF ANTARCTIC COASTAL WATERS: A ROLE FOR BOTH WINDS AND FRESHENING: AMERICAN GEOPHYSICAL UNION OCEAN SCIENCES CONFERENCE, Portland, Oregon, USA.
- Dec 2017: LOCALIZED RAPID WARMING OF WEST ANTARCTIC SUBSURFACE WATERS BY REMOTE WINDS: American Geophysical Union Fall Meeting, New Orleans, Louisiana, USA.
- Nov 2017: PHYSICAL MECHANISMS OF SEA LEVEL VARIATIONS IN A CHANGING CLIMATE: International CLIVAR Scientific Steering Group meeting, Indian Institute of Tropical Meteorology, Pune, India.
- Jul 2017: LOCALIZED RAPID WARMING OF WEST ANTARCTIC SUBSURFACE WATERS BY REMOTE WINDS: WCRP Conference on Regional Sea-level Changes and Coastal Impacts, Columbia University, New York City, USA.
- May 2017: LOCALIZED RAPID WARMING OF WEST ANTARCTIC SUBSURFACE WATERS BY REMOTE WINDS: RRS JC Ross research cruise JR16005 to Orkney Passage, Southern Ocean.
- Jan 2017: THE OCEAN MESOSCALE: OBSERVATIONS, THEORY, AND MODELING: Banff International Research Station (BIRS) workshop: *Transport in unsteady flows: From deterministic structures to stochastic models and back again*, Banff, Canada.
- July 2016: ELEMENTS OF SEA LEVEL IN A CHANGING CLIMATE: Indian Institute of Tropical Meteorology, Pune, India.
- May 2016: ELEMENTS OF SEA LEVEL IN A CHANGING CLIMATE: University of New South Wales, Sydney, Australia & Australian National University, Canberra, Australia.
- Jan 2016: ELEMENTS OF SEA LEVEL IN A CHANGING CLIMATE: Louisiana State University Chemical Engineering Department, Baton Rouge, Louisiana, USA.
- Oct 2015: IMPACTS ON OCEAN HEAT FROM THE MESOSCALE: Lamont-Doherty Earth Observatory / Columbia University, USA.

- Oct 2015: IMPACTS ON OCEAN HEAT FROM THE MESOSCALE: Stony Brook Marine Sciences, Stony Brook, USA.
- Oct 2014: IMPACTS ON OCEAN HEAT FROM THE MESOSCALE: Meeting on ocean heat uptake at the National Oceanography Centre, Southampton, UK.
- Jun 2014: IMPACTS ON OCEAN HEAT FROM THE MESOSCALE: University of Stockholm, Sweden.
- Feb 2013: SEA LEVEL IN A SUITE OF FORCED GLOBAL OCEAN-ICE SIMULATIONS: CLIVAR workshop on Sea-Level Rise, Ocean/Ice-Shelf Interactions, and Ice Sheets, Hobart, Australia
- Jan 2013: OCEAN MODEL NUMERICS AND PHYSICS: CHALLENGES FOR MESOSCALE EDDYING GLOBAL CLIMATE SIMULATIONS: 10th annual meeting of the Drakkar Ocean Modelling Consortia, Grenoble, France
- Sep 2012: SEA LEVEL IN OCEAN CLIMATE MODELS: FUNDAMENTALS AND PRACTICES: University of Tasmania, Hobart, Australia
- Mar 2011: DYNAMIC SEA LEVEL, STATIC SEA LEVEL, AND THE NON-BOUSSINESQ STERIC EFFECT: Australia National University, Canberra, Australia
- Sep 2010: SENSITIVITY OF ATLANTIC OCEAN VARIABILITY TO OCEAN PHYSICS AND VERTICAL COORDINATE: CLIVAR WGOMD/GSOP Workshop on Decadal Variability, Predictability, and Predictions: Understanding the Role of the Ocean. Boulder USA

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### INVITED PEDAGOGICAL LECTURES AND SPECIAL TOPICS COURSES

- Jun 2026: FUNDAMENTALS OF OCEAN MODELS AND THE ANALYSIS OF OCEAN SIMULATIONS. Five lectures (two hours each) concerning the ocean equations forming the foundations of NEMO. Paris, France.
- May 2026: A PRIMER ON THE OCEAN MESOSCALE AND ITS PARAMETERIZATION IN CIRCULATION MODELS: SAMPLING THE PHYSICS/MATHS/NUMERICS (WITH A FOCUS ON TRACERS), talk given at the Institut Henri Poincaré workshop: *Instabilities and Transitions in Geophysical Flows*.
- Sep 2024: THE LONG AND WINDING ROAD TO OMIP, talk given at the 50th anniversary of the Ocean Section of NCAR, National Center of Atmospheric Research, Boulder, USA.
- Oct 2022: FUNDAMENTAL EQUATIONS AND DIAGNOSTICS FOR MOM6. Lecture given as part of a 2-day tutorial on the Modular Ocean Model (MOM6). Princeton, USA.
- Feb 2022: MEDITATION FOR SCIENTISTS (with Jonathan Lilly): AMERICAN GEOPHYSICAL UNION OCEAN SCIENCES CONFERENCE. Honolulu, Hawaii (virtual).
- Feb 2020: MEDITATION FOR SCIENTISTS (with Jonathan Lilly): AMERICAN GEOPHYSICAL UNION OCEAN SCIENCES CONFERENCE, San Diego, CA.
- April/May 2019: FUNDAMENTALS OF OCEAN MODELS AND THE ANALYSIS OF OCEAN SIMULATIONS. 15 lectures (45 minutes each) on ocean model fundamentals and analysis methods given as part of the **Advanced Ocean Modelling Summer School**. Tasmania, Australia.
- Jan 2019: A HISTORICAL SURVEY OF NEUTRAL DIFFUSION METHODS AND COMMENTS ON CURRENT RESEARCH, presented during the celebration of Peter Gent's career, NCAR, Colorado, USA.
- Jan 2019: OCEAN CIRCULATION AS A PROBLEM IN MATHEMATICAL & COMPUTATIONAL PHYSICS: A HISTORICAL AND CONTEMPORARY PERSPECTIVE. Public lecture given as part of the Australian Mathematics Science Institute (AMSI) Summer School at the University of New South Wales, Sydney, Australia.
- July 2016: OCEAN MODELLING: AN INTRODUCTION FOR MATHEMATICAL PHYSICISTS: Department of Mathematics, Savitribai Phule Pune University, Pune, India.
- Jul 2016: OCEAN MODELLING AND SEA LEVEL ANALYSIS: three lectures (two hours each) at the International Centre for Theoretical Physics / Indian Institute for Tropical Meteorology: ADVANCED SCHOOL ON EARTH SYSTEM MODELLING, Pune, India
- Apr 2014: PROBLEMS AND PROSPECTS WITH OCEAN MESOSCALE EDDYING CLIMATE MODELS: Nansen Medal lecture at the European Geosciences Union annual meeting, Vienna, Austria.
- Apr 2014: PROBLEMS AND PROSPECTS WITH OCEAN MESOSCALE EDDYING CLIMATE MODELS: lecture given at a CLIVAR workshop on eddying ocean climate models, Kiel, Germany.

- Sep 2013: PROBLEMS AND PROSPECTS OF MODEL COMPARISONS: AN OCEAN PROCESS PERSPECTIVE: lecture given at a symposium celebrating the 80th birthday of Gerold Siedler, Kiel, Germany.
- Aug 2013: OCEAN MODELS AND OCEAN MODELING: LECTURES ON THE FUNDAMENTALS AND PRACTICES: Five lectures (two hours each) at the International Centre for Theoretical Physics School: FUNDAMENTALS OF OCEAN CLIMATE MODELING AT GLOBAL AND REGIONAL SCALES, Hyderabad, India
- Sep 2012: OCEAN MODELLING WITH MOM AND ITS RELATION TO AUSTRALIAN OCEAN CLIMATE SCIENCE: Second meeting of Consortia for Ocean Modelling in Australia, Hobart, Australia
- Feb 2012: OCEAN MODELLING WITH MOM AND ITS RELATION TO AUSTRALIAN OCEAN CLIMATE SCIENCE: First meeting of Consortia for Ocean Modelling in Australia, Hobart, Australia
- Nov 2010: OCEAN CLIMATE MODELING AT GFDL: Scientific Workshop for the Centre for Australian Weather and Climate Research, Hobart, Australia
- Mar 2009: PHYSICAL PROCESSES SETTING THE OCEAN'S WATER MASSES: four lectures (two hours each) at the Université Catholique de Louvain, Belgium
- Nov 2007: OCEAN MODEL FUNDAMENTALS: 10 lectures (two hours each) at the University of Tasmania, Australia
- Aug 2006: OCEAN MODEL FUNDAMENTALS: two lectures (one hour each) at the NSF summer school, MODERN MATHEMATICAL METHODS IN PHYSICAL OCEANOGRAPHY, Breckenridge, USA
- Oct 2004: OCEAN MODEL FUNDAMENTALS: 10 lectures (two hours each) at the INDIAN INTENSIVE SCHOOL ON LARGE-SCALE OCEAN MODELLING, Bangalore, India
- Sep 2004: OCEAN MODEL FUNDAMENTALS: three lectures (two hours each) at the GLOBAL OCEAN DATA ASSIMILATION EXPERIMENT SUMMER SCHOOL, La Londe Les Maures, France
- May 2003: OCEAN CLIMATE MODELING AT NOAA-GFDL: two lectures (one hour each) for a workshop on ocean modeling, Hobart, Australia
- May 2002: OCEAN CLIMATE MODELING WITH MOM4: three lectures (one hour each) for a workshop on ocean modeling, Kiel, Germany
- Jan 2001: OCEAN DYNAMICS AND MODELING: three lectures (two hours each) at La Escuela de Verano de Universidad de Concepción, Chile
- Mar 1999: OCEAN AND CLIMATE MODELING: two lectures (90 minutes each) at CONFERENCE ON GLOBAL CLIMATE, Barcelona, Spain

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## DOCUMENTS UNDER REVIEW OR IN PREPARATION

7. [Geophysical Fluid Mechanics](#), 2026: S.M. Griffies, *multi-volume book project in preparation*
6. Regional estimates of oceanic excess temperature change, 2026: E. McDonagh, A. J. G. Nurser, Q. Wu, J. Gregory, S. M. Griffies, T. W. N. Haine, J. Church, L. Clément, H. Graven, H. Hewitt, Y. Kostov, A. Marzocchi, M.-J. Messias, E. Newsom, T. Sohail, C. Turner, L. Zanna, J. Zika, *in prep for Journal of Climate*
5. Numerical Mixing in Finite Volume Ocean Models Part 1: Derivation and implementation of diagnostic formula, 2026: J. D. Zika, J. I. Bisits, H. F. Drake, R. M. Holmes, S. M. Griffies, A. H. Gibson, A. McG. Hogg, J.-E. Tesdal, *in prep for Journal of Advances in Modeling Earth Systems (JAMES)*.
4. Model sensitivity to the vertical distribution and latent heat loss from Antarctic meltwater, 2026: P. Colombo, M. H. England, C. K. Yung, H. R. S. Dawson, F. Boeira Dias, A. K. Morrison, A. E. Kiss, P. Spence, A. McC. Hogg, B. K. Galton-Fenzi, and S. M. Griffies, *submitted to JAMES*, <https://essopenarchive.org/doi/full/10.22541/essoar.15007361/v1>
3. Time averaging for tendency budget analysis, 2026: C. Bladwell, R.M. Holmes, J.D. Zika, A. E. Kiss, and S.M. Griffies, *submitted to Geoscientific Model Development*. <https://editor.copernicus.org/EGUsphere/ms.records/egusphere-2026-3672>
2. Steric sea-level response to Antarctic freshwater forcing, 2026: J. Milward, R. L. Beadling, J. P. Krasting, J. Steinberg, S. M. Griffies, J.-E. Tesdal, G. M. MacGilchrist, T. Martin, and J. MacIsaac, *in revision at Journal of Geophysical Research: Oceans*
1. The contribution of stericodynamic changes to the uncertainty of regional sea-level projections over the 21st century, 2026: J.-E. Tesdal, J. P. Krasting, R. E. Kopp, T. H. J. Hermans, P. Kumar, W. V. Sweet, S. M. Griffies, *in revision at AGU Advances*, <https://doi.org/10.22541/essoar.174>

## PEER-REVIEWED PUBLICATIONS

### 2027

181. Ocean energy pathways: a multi-scale perspective, 2027: H. Aluie, R. Barkan, S.M. Griffies, D. Marshall, B. Storer, *accepted by Annual Review of Fluid Mechanics*

### 2026

180. Steric-setup effects on US East Coast sea-level changes, 2026: J. Steinberg, S. M. Griffies, and C. Piecuch, *Geophysical Research Letters*, doi:10.1029/2026GL124322
179. Forced-dissipative two-layer quasi-geostrophic waves, instability, and turbulence over an aligned sloping bottom, 2026: M. Lobo and S. M. Griffies, *Journal of Physical Oceanography*, doi:10.1175/JPO-D-25-0205.1
178. Publishing with purpose: what AGU journals stand for, 2026: I. R. Santos, M. A. Xenopoulos, L. M. Beal, S. M. Griffies, A. Montanari, E. Camporeale, G. Destouni, J. Dixon, N. Ganushkina, A. Schubnel, H. Nguyen, D. Buczkowski, A. Hendrix, J. C. Kinney, A. Rowan, G. Caprarelli, A. Bracco, G. Viola, Q. Duan, K. Semmens, B. A. Carter, K. Lajtha, S. J. Feakins, K. Karnauskas, Y. Cheng, and K. Caylor, *Geophysical Research Letters*, doi: 10.1029/2026GL125704
177. Antarctic sea ice trends across high-resolution coupled model hierarchies, 2026: M. Bushuk, D. B. Bonan, S. M. Griffies, M. Winton, W. Gregory, Y. Zhang, B. Hurlin, and Y.-T. Chen, *Geophysical Research Letters*, doi:10.1029/2026GL124043
176. A WENO finite-volume scheme for the conservation law of potential vorticity in isopycnal ocean models, 2025: W. Zhang, Y.-H. Kuo, S. Silvestri, A. Adcroft, R.W. Hallberg, S.M. Griffies, *Journal of Advances in Modeling Earth Systems (JAMES)*, **18**, doi:10.1029/2025MS005376
175. Assessing Earth's energy imbalance trend in the early 21st century in two high-resolution coupled models, 2026: Y. T. Chen, T. M. Merlis, T. Dinh, S. M. Griffies, J. Krasting, R. Dussin, N. Zadeh, and S. A. Fueglistaler, *Geophysical Research Letters*, **53**, e2025GL121277, <https://doi.org/10.1029/2025GL121277>
174. Large regional differences in Antarctic ice shelf mass loss from Southern Ocean warming and meltwater feedbacks, 2026: M. Muilwijk, T. Hattermann, R. Beadling, N. C. Swart, A. Nummelin, C. Guo, D. M. Chandler, P. Langebroek, S. Zhou, P. Dutrieux, J.-J. Chen, C. Danek, M. J. England, S. M. Griffies, F. A. Haumann, A. Jüling, O. Jouet, Q. Li, T. Martin, J. Marshall, A. G. Pauling, A. Purich, Z. Song, I. J. Smith, Max Thomas, I. Trombini, E. van der Linden, X. Xu, *The Cryosphere*, <https://doi.org/10.5194/tc-20-1087-2026>

### 2025

173. The GFDL-CM4X climate model hierarchy, Part I: model description and thermal properties, 2025: S.M. Griffies, A. Adcroft, R.L. Beadling, M. Bushuk, C.-Y. Chang, H.F. Drake, R. Dussin, R.W. Hallberg, W.J. Hurlin, H. Khatir, J.P. Krasting, M. Lobo, G.A. MacGilchrist, B.G. Reichl, A. Sane, O. Sergienko, M. Sonnewald, J.M. Steinberg, J.-E. Tesdel, M. Thomas, K.E. Turner, M.L. Ward, M. Winton, N. Zadeh, R. Zhang, W. Zhang, M. Zhao, *Journal of Advances in Modeling Earth Systems (JAMES)*, **17**(10), e2024MS004861, <https://doi.org/10.1029/2024MS004861>
172. The GFDL-CM4X climate model hierarchy, Part II: case studies, 2025: S.M. Griffies, A. Adcroft, R.L. Beadling, M. Bushuk, C.-Y. Chang, H.F. Drake, R. Dussin, R.W. Hallberg, W.J. Hurlin, H. Khatir, J.P. Krasting, M. Lobo, G.A. MacGilchrist, B.G. Reichl, A. Sane, O. Sergienko, M. Sonnewald, J.M. Steinberg, J.-E. Tesdel, M. Thomas, K.E. Turner, M.L. Ward, M. Winton, N. Zadeh, R. Zhang, W. Zhang, M. Zhao, *Journal of Advances in Modeling Earth Systems (JAMES)*, **17**(10), e2024MS004862, <https://doi.org/10.1029/2024MS004862>
171. A review of Green's function methods for tracer timescales and pathways in ocean models, 2025: T.W.N. Haine, S.M. Griffies, G. Gebbie, and W. Jiang, *Journal of Advances in Modeling Earth Systems (JAMES)*, **17**, doi:10.1029/2024MS004637
170. Water mass transformation budgets in finite-volume generalized vertical coordinate ocean models, 2025: H. F. Drake, S. Bailey, R. Dussin, S. M. Griffies, J. Krasting, G. MacGilchrist, G. Stanley, J.-E. Tesdal, J. D. Zika, *Journal of Advances in Modeling Earth Systems (JAMES)*, **17**, doi:10.1029/2024MS004383
169. Vertical structure of baroclinic instability in a three-layer quasi-geostrophic model over a sloping bottom, 2025: M. Lobo, S. M. Griffies, and W. Zhang, *Journal of Physical Oceanography*, **55**, 341–359, doi:10.1175/JPO-D-24-0130.1

### 2024

168. The averaged hydrostatic Boussinesq equations in generalized vertical coordinates, 2024: M. Janson, A. Adcroft, S. M. Griffies, and I. Grooms, *Journal of Advances in Modeling Earth Systems (JAMES)*, **16**, doi:10.1029/2024MS004506
167. A link between U.S. east coast sea level and North Atlantic subtropical ocean heat content, 2024: J. M. Steinberg, S. M. Griffies, J. P. Krasting, C. G. Piecuch, and Andrew C. Ross, *Journal of Geophysical Research: Oceans*, **129**, doi:10.1029/2024JC021425
166. Steric sea level rise in GFDL CM4 and ESM4: insights into model drift and water mass representation, 2024: K. P. Krasting, S. M. Griffies, J.-E. Tesdal, G. A. MacGilchrist, R. L. Beadling, and C. M. Little, *Journal of Climate*, **37**, 65636583, doi:10.1175/JCLI-D-23-0591.1
165. Improved upper ocean vertical mixing in the tropics in NOAA/GFDL's OM4 model, 2024: B. G. Reichl, A. T. Wittenberg, S. M. Griffies, A. Adcroft, *Earth and Space Science*, **11**, doi:10.1029/2023EA003485

164. A scale-dependent analysis of the barotropic vorticity budget in a global ocean simulation, 2023: H. Khatri, S. M. Griffies, B. A. Storer, M. Buzzicotti, H. Aluie, M. Sonnewald, R. Dussin, and A. Shao, *Journal of Advances in Modeling Earth Systems (JAMES)*, **16**, doi:10.1029/2023MS003813
163. The role of surface potential vorticity in the vertical structure of mesoscale eddies in wind-driven ocean circulations, 2024: W. Zhang, S. M. Griffies, R. W. Hallberg, Y.-H. Kuo, C. L. P. Wolfe, *Journal of Physical Oceanography*, doi:10.1175/JPO-D-23-0203.1
162. Ross Gyre variability modulates oceanic heat supply to the Antarctic continental shelf, 2024: C.J. Prend, G.A. MacGilchrist, G.E. Manucharyan, R.Q. Pang, R. Moorman, A.F. Thompson, S.M. Griffies, Matthew R. Mazloff, Lynne D. Talley, and Sarah T. Gille, *Science Advances*, doi:10.1038/s43247-024-01207-y
161. The role of bottom friction in mediating the response of the Weddell Gyre circulation to changes in surface stress and buoyancy fluxes, 2024: J. Neme, M. H. England, A. McC. Hogg, H. Khatri, and S. M. Griffies, *Journal of Physical Oceanography*, **54**, 216-236, doi:10.1175/JPO-D-23-0165.1

## 2023

160. A new conceptual model of global ocean heat uptake, 2023: J. M. Gregory, J.S. Bloch-Johnson, M.P. Couldrey, E. Exarchou, S.M. Griffies, T. Kuhlbrodt, E. Newsom, O.A. Saenko, T. Suzuki, Q. Wu, S. Urakawa, and L. Zanna, *Climate Dynamics*, doi:10.1007/s00382-023-06989-z
159. Global cascade of kinetic energy in the ocean and the atmospheric imprint, 2023: B.A. Storer, M. Buzzicotti, H. Khatri, S.M. Griffies, and H. Aluie, *Science Advances*, doi:10.1126/sciadv.adi7420
158. Comparing two parameterizations for the restratification effect of mesoscale eddies in an isopycnal ocean model, 2023: N. Loose, G.M. Marques, A. Adcroft, S. Bachman, S.M. Griffies, I. Grooms, R. Hallberg, and M.F. Jansen, *Journal of Advances in Modeling Earth Systems (JAMES)*, **15**, doi:10.1029/2022MS003518
157. The Southern Ocean Freshwater release model experiments Initiative (SOFIA): Scientific objectives and experimental design, 2023: N. Swart, T. Martin, R. Beadling, J.-J. Chen, M.H. England, R. Farneti, S.M. Griffies, T. Hattermann, F.A. Haumann, Q. Li, J. Marshall, M. Muilwijk, A.G. Pauling, A. Purich, I.J. Smith, and M. Thomas, *Geoscientific Model Development*, **16**, 72897309, doi:10.5194/gmd-16-7289-2023
156. Remote versus local impacts of energy backscatter on the North Atlantic SST biases in a global ocean model, 2023: C.-Y. Chang, A. Adcroft, L. Zanna, R. W. Hallberg, S. M. Griffies, *Geophysical Research Letters*, doi:10.1029/2023GL105757
155. Sensitivity of Antarctic shelf waters and abyssal overturning to local wind amplitude, 2023: A. K. Morrison, W. G. C. Huneke, J. Neme, P. Spence, A. McC. Hogg, M. H. England, and S. M. Griffies, *Journal of Climate*, doi:10.1175/JCLI-D-22-0858.1
154. Exploring the non-stationarity of coastal sea level probability distributions, 2023: F. Falasca, A. Brettin, L. Zanna, S.M. Griffies, J. Yin, and M. Zhao, *Environmental Data Science*, doi:10.1017/eds.2023.10
153. Spatio-temporal coarse-graining decomposition of the global ocean geostrophic kinetic energy, 2023: M. Buzzicotti, B. A. Storer, H. Khatri, S.M. Griffies, and H. Aluie, *Journal of Advances in Modeling Earth Systems (JAMES)*, doi:10.1029/2023MS003693
152. Revisiting interior water mass responses to surface forcing changes and the subsequent effects on overturning in the Southern Ocean, 2023: J.-E. Tesdal, G.A. MacGilchrist, R.L. Beadling, S.M. Griffies, J.P. Krasting, and P.J. Durack, *JGR-Oceans*, **128**, doi:10.1029/2022JC019105

## 2022

151. Effective drift velocity from turbulent transport by vorticity, 2022: H. Aluie, S. Rai, H. Yin, A. Lees, D. Zhao, S.M. Griffies, A. Adcroft, and J.K. Shang, *Physical Review Fluids*, **7**, doi:10.1103/PhysRevFluids.7.104601
150. Global energy spectrum of the general oceanic circulation, 2022: B.A. Storer, M. Buzzicotti, H. Khatri, S.M. Griffies, and H. Aluie, *Nature Communications*, doi:10.1038/s41467-022-33031-3
149. Surface quasigeostrophic turbulence in variable stratification, 2022: H. Yassin and S.M. Griffies, *Journal of Physical Oceanography*, doi:10.1175/JPO-D-22-0040.1
148. On the discrete normal modes of quasigeostrophic theory, 2022: H. Yassin and S.M. Griffies, *Journal of Physical Oceanography*, **52**, 243–259. doi:10.1175/JPO-D-21-0199.1
147. Importance of the Antarctic Slope Current in the Southern Ocean response to ice sheet melt and wind stress change, 2022: R.L. Beadling, J.P. Krasting, S.M. Griffies, W.J. Hurlin, B. Bronselear, J.L. Russell, G. A. MacGilchrist, J.-E. Tesdal, M. Winton, *JGR-Oceans*, **127**, e2021JC017608, doi:10.1029/2021JC017608
146. Kinetic energy transfers between mesoscale and submesoscale motions, 2022: A.C. Naveira Garabato, X. Yu, J. Callies, R. Barkan, K.L. Polzin, E.E. Frajka-Williams, C.E. Buckingham, and S.M. Griffies, *Journal of Physical Oceanography*, **52**, 75–97, doi:10.1175/JPO-D-21-0099.1
145. Local drivers of marine heatwaves: A global analysis with an Earth system model, 2022: L. Vogt, F.A. Burger, S.M. Griffies, and T.L. Frölicher, *Frontiers in Climate*, 4:847995, doi:10.3389/fclim.2022.847995
144. NeverWorld2: An idealized model hierarchy to investigate ocean mesoscale eddies across resolutions, 2022: G.M. Marques, N. Loose, E. Yankovsky, J.M. Steinberg, C.-Y. Chang, N. Bhamidipati, A. Adcroft, B. Fox-Kemper, S.M. Griffies, R.W. Hallberg, M.F. Jansen, H. Khatri, and L. Zanna, *Geoscientific Model Development*, doi:10.5194/gmd-15-6567-2022

143. A potential energy analysis of ocean surface mixed layers, 2022: B. Reichl, A. Adcroft, S.M. Griffies, and R.W. Hallberg, *JGR-Oceans*, **127**, e2021JC018140, doi:10.1029/2021JC018140
142. Effects of grid spacing on high-frequency precipitation variance in coupled high-resolution global ocean-atmosphere models, 2022: C.X. Light, B.K. Arbic, P.E. Martin, L. Brodeau, J.T. Farrar, S.M. Griffies, B.P. Kirtman, L.C. Laurindo, D. Menemenlis, A. Molod, A.D. Nelson, E. Nyadjro, A.K. O'Rourke, J.F. Shriver, L. Siqueira, R.J. Small, and E. Strobach, *Climate Dynamics*, doi:10.1007/s00382-022-06257-6

## 2021

141. The interpretation of temperature and salinity variables in numerical ocean model output, and the calculation of heat fluxes and heat content, 2021: T.J. McDougall, P.M. Barker, R.M. Holmes, R. Pawlowicz, S.M. Griffies, and P.J. Durack, *Geoscientific Model Development*, **14**, 64456466, doi:10.5194/gmd-14-6445-2021
140. Role of mixed-layer instabilities in the seasonal evolution of eddy kinetic energy spectra in a global submesoscale permitting simulation, 2021: H. Khatri, S.M. Griffies, T. Uchida, D. Menemenlis, and H. Wang, *Geophysical Research Letters*, e2021GL094777, doi:10.1029/2021GL094777
139. The geography of numerical mixing in a suite of global ocean models, 2021: R.M. Holmes, J.D. Zika, S.M. Griffies, A.McC. Hogg, A.E. Kiss, and M.H. England, *Journal of Advances in Modeling Earth Systems (JAMES)*, **13**, doi:10.1029/2020MS002333
138. A mechanistic analysis of tropical Pacific dynamic sea level in GFDL-OM4 under OMIP-I and OMIP-II forcings, 2021: C.-W. Hsu, J. Yin, S.M. Griffies, and R. Dussin, *Geoscientific Model Development*, **14**, 2471–2502, doi:10.5194/gmd-14-2471-2021
137. On the role of the Antarctic Slope Front on the occurrence of the Weddell Sea polynya under climate change, 2021: J.W. Lockwood, C.O. Dufour, S.M. Griffies, and M. Winton, *Journal of Climate*, doi:10.1175/JCLI-D-20-0069.1
136. What causes the spread of model projections of ocean dynamic level change in response to greenhouse gas forcing?, 2021: M.P. Couldrey, J.M. Gregory, F. Boeira Dias, P. Dobrohotoff, C. Domingues, O. Garuba, S.M. Griffies, H. Haak, A. Hu, M. Ishii, J. Jungclaus, A. Köhlaffil, S. Marsland, S. Ojha, O.A. Saenko, A. Savita, A. Shao, D. Stammer, T. Suzuki, A. Todd, L. Zanna, *Climate Dynamics*, doi:10.1007/s00382-020-05471-4

## 2020

135. Contribution of ocean physics and dynamics at different scales to heat uptake in low-resolution AOGCMs, 2020: O. Saenko, J.M. Gregory, S.M. Griffies, and F. Boeira Dias, *Journal of Climate*, doi:10.1175/JCLI-D-20-0652.1
134. A general-coordinate, non-local neutral diffusion operator, 2020: A. Shao, A. Adcroft, R.W. Hallberg, and S.M. Griffies, *Journal of Advances in Modeling Earth Systems (JAMES)*, doi:10.1029/2019MS001992
133. A primer on ocean generalized vertical coordinate dynamical cores based on the vertical Lagrangian-remap method, 2020: S.M. Griffies, A.J. Adcroft, and R.W. Hallberg, *Journal of Advances in Modeling Earth Systems (JAMES)*, **12**, doi:10.1029/2019MS001954
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